

VARUN JOSE MADANU

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PROFESSIONAL SUMMARY

AI/ML Engineer with 4+ years across research-engineering and production software, holding an M.S. in Computer Science and hands-on experience authoring and reviewing evaluation tasks for frontier language models. Strong Python scripting and debugging skills with clean-code habits, everyday fluency with Git, IDEs, and standard development workflows, and daily use of AI coding assistants and agent frameworks. Experienced in designing multi-step engineering tasks with verifiable reference solutions, diagnosing where and why AI agents fail, and communicating findings clearly to researchers. Works independently through ambiguous, open-ended problems with high attention to detail.

PROFESSIONAL EXPERIENCE

AI/ML Evaluation Analyst (Contract), Handshake AI | Remote, USA | 03/2026 – 07/2026

- Authored multi-step software engineering evaluation tasks for frontier language models, designing realistic problems drawn from day-to-day engineering work that current AI coding agents could not solve reliably end to end.
- Built verifiable reference solutions in Python for each authored task, including environment setup, dependency pinning, fixtures, and automated checks so every task had an unambiguous pass/fail signal.
- Reviewed tasks submitted by fellow contributors for clarity, correctness, difficulty calibration, and reproducibility, providing written feedback that reduced ambiguity and rework in downstream grading.
- Analyzed agent transcripts and failure traces to categorize recurring breakdown modes misread requirements, incorrect tool use, premature stopping, silent test failures and summarized findings for research stakeholders.
- Used AI coding assistants as part of the daily workflow while authoring and solving tasks, documenting where assistance accelerated work and where it produced plausible but incorrect output.
- Maintained detailed written documentation and task specifications, working independently against open-ended prompts and delivering consistently on a part-time weekly commitment alongside a full-time role.

AI/ML Engineer, Fiserv | Remote, USA | 09/2024 – 02/2026

- Designed and shipped an agentic AI investigation system from an ambiguous problem statement to production, building an agent harness with LangGraph, LangChain, MCP servers, and Retrieval-Augmented Generation for autonomous tool use, evidence collection, and explainable decisioning increasing investigator productivity by 34%.
- Built the evaluation infrastructure around it: CI-gated evaluation suites, regression tests, and benchmark reporting that measured agent quality, latency, and cost before any prompt or model change merged.
- Established experimental rigor and observability for production systems shadow deployments, A/B testing, drift detection, and adversarial robustness evaluation improving production reliability by 27%.
- Fine-tuned Llama 3.1 and Mistral 7B using LoRA, QLoRA, Direct Preference Optimization, and Bayesian optimization, reproducing recent post-training research and converting it into reusable internal tooling.
- Wrote and maintained production Python services and FastAPI APIs on UNIX infrastructure with pytest coverage, code review, and documentation standards, using agentic coding tools to accelerate delivery.
- Architected AWS-native streaming pipelines (MSK, Kinesis, Lambda, Glue, S3, EMR, Neptune Analytics, Redshift) with entity resolution, schema validation, and data lineage, reducing investigation latency by 31%.
- Shipped a production graph intelligence platform using GraphSAGE, GATv2, and Temporal Graph Networks to detect payment fraud, mule networks, and merchant abuse, improving detection accuracy by 29%.

ML Engineer, S&P Global India Pvt. Ltd. | Hyderabad, India | 05/2021 – 07/2023

- Built and operated distributed data infrastructure with Apache Spark, Kafka, Airflow, and SQL, processing millions of financial records daily with standardized schemas and automated validation.
- Developed real-time anomaly detection and risk intelligence systems over market data, pricing feeds, and trading events, improving incident detection speed by 28%.
- Trained, benchmarked, and optimized production ML models (XGBoost, LightGBM, Autoencoders) for credit risk prediction and pricing validation, improving predictive accuracy by 32%.
- Applied BERT and TF-IDF with logistic regression to financial news and analyst research, improving document classification and prioritization efficiency by 42%.

- Containerized ML services with Docker on Kubernetes for real-time inference, reducing inference latency by 38%.
- Built ML experimentation frameworks and monitoring dashboards tracking pipeline health, feature quality, model performance, and drift for engineering and product teams.

TECHNICAL SKILLS

- **Python & Software Engineering:** Python (scripting, debugging, profiling), pytest, type hints, clean-code and readability practices, Bash, UNIX/Linux, Git & GitHub, code review, IDEs (VS Code, PyCharm), virtual environments, Docker, dependency and environment setup, REST APIs, FastAPI, technical writing and documentation
- **AI Evaluation & Benchmarking:** LLM evaluation harnesses, benchmark and task authoring, rubric design, reference solution construction, automated graders and verifiable checks, failure-mode analysis, inter-rater review, regression testing, CI-gated evaluations, red-teaming, adversarial robustness evaluation, human-in-the-loop review
- **AI Coding Assistants & Agent Workflows:** Claude Code, Cursor, GitHub Copilot, prompt engineering, ReAct, self-reflection, tool-use systems, agent harnesses, LangGraph, LangChain, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), Claude API, OpenAI API, Hugging Face Transformers, Llama, Mistral
- **ML, Deep Learning & Post-Training:** PyTorch, TensorFlow, Scikit-learn, Transformers, XGBoost, LightGBM, CatBoost, Autoencoders, Graph Neural Networks (GraphSAGE, GATv2), Temporal Graph Networks, Supervised Fine-Tuning, LoRA, QLoRA, Direct Preference Optimization (DPO), Bayesian optimization, Explainable AI (SHAP)
- **Data, MLOps & Cloud:** SQL, PostgreSQL, Redis, Pandas, NumPy, Apache Spark, Kafka, Airflow, ETL pipelines, MLflow, SageMaker Pipelines, Kubernetes, GitHub Actions, CI/CD, observability and logging, AWS (SageMaker, Lambda, S3, Glue, EMR, Kinesis, MSK, Redshift, Neptune Analytics), GCP (Vertex AI, Gemini), Vector DBs (FAISS, Pinecone, ChromaDB)

PROJECTS

Multi-Agent RAG Platform with Evaluation Harness

- Built a planner-specialist multi-agent platform on Vertex AI with Gemini, Claude, LangGraph, FAISS, PostgreSQL, FastAPI, and Docker, implementing ReAct, self-reflection, MCP servers, and human-in-the-loop checkpoints.
- Authored the evaluation harness task suites, automated graders, CI-gated regression testing, and benchmark reporting so every agent change was scored against a fixed set before merge.

Enterprise RAG Pipeline for Client Workflow

- Delivered a production RAG pipeline using the Claude API, ChromaDB, Redis, GCP, AWS, and MCP servers, tuning chunking, latency, cost, and answer quality against a defined benchmark set.
- Added PII redaction, audit logging, versioned prompt libraries, and documentation to keep the system operable by other engineers.

Sleep Apnea Prediction Using Deep Learning

- Designed a PyTorch CNN over 3D depth maps with feature engineering, augmentation, cross-validation, and rigorous evaluation; supported research through paper submission efforts.

EDUCATION

University at Albany, SUNY — Albany, NY, USA

Master of Science (M.S.) in Computer Science | 08/2023 – 05/2025

Kommuri Pratap Reddy Institute of Technology — Hyderabad, India

Bachelor of Technology (B.Tech.) in Computer Science & Engineering | 03/2018 – 06/2022